Eigenvalues and eigenvectors

One of the most important problems in linear algebra is the *eigenvalue problem*. It asks the following question: If A is an $n \times n$ matrix, does there exist a **non-zero vector** v such that Av is a scalar multiple of v? As an illustration, let A be the 2×2 matrix

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$$

and observe that

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

If $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ then $Av_1 = (1)v_1$, thus Av_1 is a scalar multiple of the vector v_1. Similarly, if $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ then $Av_2 = (-1)v_2$ and thus Av_2 is also a scalar multiple of v_2. Note that

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \pi \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \sqrt{3} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \cdots$$

Thus, if $\Theta = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then $A\Theta = \lambda\Theta$ is always true for any real number λ. The eigenvalue problem for the 2×2 case asks if there is a vector $v \neq \Theta$ which satisfies $Av = \lambda v$. We have obtained two of them for the matrix A given above.

Definition of eigenvalue and eigenvector

> **Definition 1.** Let A be an $n \times n$ matrix. A number λ is called an eigenvalue of A provided there exists a non-zero vector v in $\mathbb{R}^n$ such that
>
> $$Av = \lambda v$$
>
> Every non-zero vector satisfying this equation is called an *eigenvector A corresponding to the eigenvalue λ.*

Example 1. Find the eigenvalue and eigenvectors of the following matrix

$$A = \begin{pmatrix} 2 & -12 \\ 1 & -5 \end{pmatrix}$$

The steps are as follows.

- Find the determinant of $A - \lambda I$, where I is the 2×2 identity matrix:

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -12 \\ 1 & -5 - \lambda \end{vmatrix} = \lambda^2 + 3\lambda + 2$$

- The polynomial that we get when we simplify $\det(A - \lambda I)$ is called the *characteristic polynomial*.

  When we equate it to zero, we obtain the *characteristic equation*. Solve the characteristic equation to get the values of λ:

$$\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0 \Rightarrow \lambda_1 = -1 \text{ and } \lambda_2 = -2.$$

- The roots of the characteristic equation are the eigenvalues of A. In the present case, they are $\lambda_1 = -1$ and $\lambda_2 = -2$.

- To find the first eigenvector, we solve homogeneous system of equation with the equation with $(A - \lambda_1 I)$ as the coefficient matrix:

$$\left[\begin{pmatrix} 2 & -12 \\ 1 & -5 \end{pmatrix} - (-1)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right]\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & -12 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

  Gauss-elimination method gives the solution set $S = \{t(4, 1)|t \in \mathbb{R}\}$. Now, choose any value of t to obtain a particular non-zero vector in S which will be the eigenvector for $\lambda_1 = -1$. For example, for $t = 1$, the eigenvector is $v_1 = (4, 1)$.

- Similarly, for $\lambda_2 = -2$, we solve the homogeneous system of equation with $A - \lambda_2 I$ as the coefficient matrix:

$$\begin{pmatrix} 4 & -12 \\ 1 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

  The solution set is $S = \{t(3, 1)|t \in \mathbb{R}\}$. A particular eigenvector, for example, is $v_2 = (6, 2)$ (for $t = 2$).

Example 2. Find the eigenvalues and eigenvectors of the following matrix

$$A = \begin{pmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{pmatrix}$$

- Solve the characteristic equation

$$\det(A - \lambda I) = \begin{vmatrix} -2 - \lambda & -4 & 2 \\ -2 & 1 - \lambda & 2 \\ 4 & 2 & 5 - \lambda \end{vmatrix} = (\lambda - 3)(\lambda + 5)(\lambda - 6) = 0$$

  The eigenvalues are the roots $\lambda_1 = 3$, $\lambda_2 = -5$ and $\lambda_3 = 6$.

- For $\lambda_1 = 3$, solve the system of equation

$$\begin{pmatrix} -5 & -4 & 2 \\ -2 & -2 & 2 \\ 4 & 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  to obtain a solution $(2, -3, -1)$. Thus the eigenvector corresponding to the eigenvalue $\lambda_1 = 3$ is $v_1 = (2, -3, -1)$

- For $\lambda_2 = -5$, solve the system of equation

$$\begin{pmatrix} 3 & -4 & 2 \\ -2 & 6 & 2 \\ 4 & 2 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  to obtain a solution $(2, -1, 1)$. Thus the eigenvector corresponding to the eigenvalue $\lambda_2 = -5$ is $v_2 = (2, -1, 1)$

- For $\lambda_3 = 6$, solve the system of equation

$$\begin{pmatrix} -8 & -4 & 2 \\ -2 & -5 & 2 \\ 4 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  to obtain a solution $(1, 6, 16)$. Thus the eigenvector corresponding to the eigenvalue $\lambda_3 = 6$ is $v_3 = (1, 6, 16)$.

The following two examples show how to find eigenvectors in case a eigenvalue is a repeated root of the characteristic equation.

Example 3. Find the eigenvalues and eigenvectors of the following matrix

$$A = \begin{pmatrix} 4 & 6 & 0 \\ -3 & -5 & 0 \\ -3 & -6 & 1 \end{pmatrix}$$

- Solve the characteristic equation

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 6 & 0 \\ -3 & -5-\lambda & 0 \\ -3 & -6 & 1-\lambda \end{vmatrix} = (\lambda - 1)^2(\lambda + 2) = 0$$

  The eigenvalues are the roots $\lambda_1 = \lambda_2 = 1$ (repeated) and $\lambda_3 = -2$.

- For $\lambda_1 = \lambda_2 = 1$, solve the system of equation

$$\begin{pmatrix} 3 & 6 & 0 \\ -3 & -6 & 0 \\ -3 & -6 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  to obtain the solution set $S = \{s(2, -1, 0) + t(0, 0, 1) | s, t \in \mathbb{R}\}$. The two eigenvectors are obtained from S by substituting $s = 0, t = 1$ and $s = 1, t = 0$. Thus we get $v_1 = (0, 0, 1)$ and $v_2 = (2, -1, 0)$.

- For $\lambda_3 = -2$, solve the system of equation

$$\begin{pmatrix} 6 & 6 & 0 \\ -3 & -3 & 0 \\ -3 & -6 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  to obtain the solution set $S = \{s(-1, 1, 1) | s \in \mathbb{R}\}$. The eigenvector corresponding to the eigenvector $\lambda_3 = -2$ is obtained from S by substituting, for example, $s = 1$, that is $v_3 = (-1, 1, 1)$.

Example 4. Find the eigenvalues and eigenvectors of the following matrix

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{pmatrix}$$

- Solve the characteristic equation

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & 0 & 0 \\ 0 & 1-\lambda & 5 & -10 \\ 1 & 0 & 2-\lambda & 0 \\ 1 & 0 & 0 & 3-\lambda \end{vmatrix} = (\lambda - 1)^2(\lambda - 2)(\lambda - 3) = 0$$

  The eigenvalues are the roots: $\lambda_1 = 1$, $\lambda_2 = 1$ (1 is a repeated root), $\lambda_3 = 2$ and $\lambda_4 = 3$.

- For $\lambda_1 = \lambda_2 = 1$, solve the system

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & -10 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

  to get the solution set $S = \{s(0,1,0,0) + t(-2,0,2,1) | s,t \in \mathbb{R}\}$. To get two eigenvectors corresponding to the eigenvector $\lambda_1 = \lambda_2 = 1$, we put $s = 0, t = 1$ and $s = 1, t = 0$ to obtain $v_1 = (-2,0,2,1)$ and $v_2 = (0,1,0,0)$.

- For $\lambda_3 = 2$, solve

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 5 & -10 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

  to get the solution set $S = \{s(0,5,1,0) | s \in \mathbb{R}\}$. Hence an eigenvector for $\lambda_3 = 2$ is $v_3 = (0,5,1,0)$.

- For $\lambda_4 = 3$, solve

$$\begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 5 & -10 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

  to get the solution set $S = \{s(0,-5,0,1) | s \in \mathbb{R}\}$. Hence an eigenvector for $\lambda_4 = 3$ is $v_4 = (0,-5,0,1)$.

Properties of eigenvalues and eigenvectors

We list some of the important properties of the eigenvalues and eigenvectors of any $n \times n$ matrix A.

1. The sum of the eigenvalues of A is equal to the sum of the entries in the main diagonal of A.
2. The product of the eigenvalues of A is equal to the determinant of A.
3. The matrix A and its transpose A^T have the same eigenvalues.
4. If λ is an eigenvalue of A, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}. (If A^{-1} exists, then $\det(A) \neq 0$. Hence by the second property, no eigenvalue λ of A can be zero.)
5. If v is an eigenvector of A corresponding to the eigenvalue λ, then λ^n is an eigenvector of A^n and the corresponding eigenvector is v.

The following is a simple application of the above properties.

Example 5. The product of two eigenvalues of the matrix $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ is 16. Find the third eigenvalue.

Let the eigenvalues be λ_1, λ_2 and λ_3. We know that $\lambda_1\lambda_2\lambda_3 = \det(A) = 32$. Hence, $16\lambda_3 = 32$ which gives the third eigenvalue as $\lambda_3 = 2$.

Exercise

1. Find the eigenvalues and eigenvectors of the following matrices

$$A_1 = \begin{pmatrix} -3 & 2 & 3 \\ -1 & -2 & 1 \\ -3 & 2 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 2 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

2. Two of the eigenvalues of $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ are 2 and 8. Find the third eigenvalue.
3. Two of the eigenvalues of $A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$ are 3 and 6. Find the eigenvalues of A^{-1}.